EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. If $\lim\limits_{x\to a}\dfrac{2x-\sqrt{x^2+3a^2}}{\sqrt{x+a}-\sqrt{2a}}=\sqrt{2}$ (where $a\in\mathbb{R}^+$), then a is equal to -

(A) $\dfrac{1}{3}$ (B) $\dfrac{1}{2\sqrt{2}}$ (C) $\dfrac{1}{3\sqrt{2}}$ (D) $\dfrac{1}{9}$

MLT217

2. $\lim\limits_{x\to 0}\dfrac{\ell n(\sin 3x)}{\ell n(\sin x)}$ is equal to

(A) 0 (B) 1 (C) 2 (D) Non existent

MLT218

3. $\lim\limits_{x\to 0}\dfrac{\sqrt[3]{1+x^2}-\sqrt[4]{1-2x}}{x+x^2}$ is equal to

(A) $\dfrac{1}{4}$ (B) $\dfrac{1}{2}$ (C) 1 (D) D.N.E.

MLT219

4. $\lim\limits_{x\to 1}\dfrac{\sqrt[3]{7+x^3}-\sqrt{3+x^2}}{x-1}$ is equal to

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(A) $\dfrac{1}{4}$ (B) $\dfrac{1}{6}$ (C) $-\dfrac{1}{4}$ (D) $-\dfrac{1}{6}$

MLT220

5. $\lim\limits_{x\to\infty}\dfrac{(x+1)^{10}+(x+2)^{10}+....+(x+100)^{10}}{x^{10}+10^{10}}$ is equal to

(A) 1 (B) 100 (C) 200 (D) 10

MLT221

6. $\lim\limits_{x\to-\infty}\left(\sqrt{x^2-2x-1}-\sqrt{x^2-7x+3}\right)$ is equal to

(A) $-\dfrac{5}{2}$ (B) $\dfrac{5}{2}$ (C) 0 (D) D.N.E

MLT222

7. If $\lim\limits_{n\to\infty}\left(\sqrt{2n^2+n}-\lambda\sqrt{2n^2-n}\right)=\dfrac{1}{\sqrt{2}}$ (where λ is a real number), then-

(A) $\lambda = 1$ (B) $\lambda = -1$ (C) $\lambda = \pm 1$ (D) $\lambda\in(-\infty, 1)$

MLT223

8. Let $U_n = \dfrac{n!}{(n+2)!}$ where $n\in\mathbb{N}$. If $S_n = \sum\limits_{r=1}^{n}U_r$ then $\lim\limits_{n\to\infty} S_n$ equals

(A) 2 (B) 1 (C) $\dfrac{1}{2}$ (D) Non existent

MLT224

For More Join: @IITJEE_Advanced

9. For $n \in \mathbb{N}$, let $a_n = \sum_{k=1}^{n} 2k$ and $b_n = \sum_{k=1}^{n} (2k-1)$. Then $\lim_{n\to\infty}\left(\sqrt{a_n} - \sqrt{b_n}\right)$ is equal to-

(A) 1 (B) $\frac{1}{2}$ (C) 0 (D) 2

MLT225

10. Let $P_n = \prod_{k=2}^{n}\left(1 - \frac{1}{{}^{k+1}C_2}\right)$. If $\lim_{n\to\infty} P_n$ can be expressed as lowest rational in the form $\frac{a}{b}$, then value of $(a+b)$ is

(A) 4 (B) 8 (C) 10 (D) 12

MLT226

11. $\lim_{x\to -1} \frac{\cos 2 - \cos 2x}{x^2 - |x|}$ is equal to

(A) 0 (B) cos2 (C) 2sin2 (D) sin1

MLT227

12. $\lim_{x\to 0}\left(\left[\frac{-5\sin x}{x}\right] + \left[\frac{6\sin x}{x}\right]\right)$ (where [.] denotes greatest integer function) is equal to -

(A) 0 (B) –12 (C) 1 (D) 2

MLT228

13. Let $f(x) = \left[\frac{\sin x}{x}\right] + \left[\frac{2\sin 2x}{x}\right] + ... + \left[\frac{10\sin 10x}{x}\right]$ (where $[y]$ is the largest integer $\le y$). The value of $\lim_{x\to 0} f(x)$ equals

(A) 55 (B) 164 (C) 165 (D) 375

MLT229

14. Let $f(x) = \frac{\sin\{x\}}{x^2 + ax + b}$. If $f(5^+)$ & $f(3^+)$ exists finitely and are not zero, then the value of $(a+b)$ is (where {.} represents fractional part function)-

(A) 7 (B) 10 (C) 11 (D) 20

MLT230

15. $\lim_{x\to 0} \frac{\left|\cos(\sin(3x))\right| - 1}{x^2}$ equals

(A) $\frac{-9}{2}$ (B) $\frac{-3}{2}$ (C) $\frac{3}{2}$ (D) $\frac{9}{2}$

MLT231

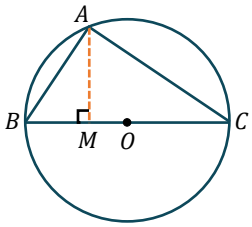
16. Let $a = \min\{x^2 + 2x + 3, x \in \mathbb{R}\}$ and $b = \lim_{\theta\to 0} \frac{1-\cos\theta}{\theta^2}$. Then value of $\sum_{r=0}^{n} a^r . b^{n-r}$ is :

(A) $\frac{2^{n+1}-1}{3.2^n}$ (B) $\frac{2^{n+1}+1}{3.2^n}$ (C) $\frac{4^{n+1}-1}{3.2^n}$ (D) $\frac{4^{n-1}-1}{3.2^n}$

MLT232

For More Join: @IITJEE_Advanced

17. Let BC is diameter of a circle centred at O. Point A is a variable point, moving on the circumference of circle. If $BC = 1$ unit, then $\lim\limits_{A\to B}\dfrac{BM}{(\text{Area of sector OAB})^2}$ is equal to -



(A) 1 (B) 2 (C) 4 (D) 16

MLT233

18. $\lim\limits_{x\to\infty}\left(\dfrac{x^2-2x+1}{x^2-4x+2}\right)^x$ is equal to

(A) 1 (B) e (C) $\dfrac{1}{e^2}$ (D) e^2

MLT234

19. $\lim\limits_{x\to 0}(1+\sin x)^{\cos x}$ is equal to

(A) 0 (B) e (C) 1 (D) $\dfrac{1}{e}$

MLT235

20. $\lim\limits_{x\to 0}(\cos x+a\sin bx)^{\frac{1}{x}}$ is equal to

(A) e^a (B) e^{ab} (C) e^b (D) $e^{a/b}$

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MLT236

21. $\lim\limits_{x\to 0}\left(\tan\left(\dfrac{\pi}{4}+x\right)\right)^{1/x}$ is equal to

(A) e^{-2} (B) $\dfrac{1}{e}$ (C) e (D) e^2

MLT237

22. $\lim\limits_{n\to\infty}(4^n+5^n)^{1/n}$ is equal to

(A) 5 (B) 4 (C) 0 (D) D.N.E.

MLT238

23. $\lim\limits_{x\to\infty}\left(\dfrac{1^{1/x}+2^{1/x}+3^{1/x}+.....+n^{1/x}}{n}\right)^{nx}$, $n\in\mathbb{N}$ is equal to

(A) $n!$ (B) 1 (C) $\dfrac{1}{n!}$ (D) 0

MLT239

24. If $\lim\limits_{x\to\lambda}\left(2-\dfrac{\lambda}{x}\right)^{\lambda\tan\left(\frac{\pi x}{2\lambda}\right)}=\dfrac{1}{e}$, then λ is equal to -

(A) $-\pi$ (B) π (C) $\dfrac{\pi}{2}$ (D) $-\dfrac{2}{\pi}$

MLT240

For More Join: @IITJEE_Advanced

25. If $\lim\limits_{x\to 0}\left(1+ax+bx^2\right)^{2/x}=e^e$, then

(A) $a=\frac{3}{2}$ and $b\in\mathbb{R}$ (B) $a=\frac{3}{2}$ and $b\in\mathbb{R}^+$

(C) $a = 0$ and $b = 1$ (D) $a = 1$ and $b = 0$

MLT241

26. If $f(x)$ is a polynomial of least degree, such that $\lim\limits_{x\to 0}\left(1+\frac{f(x)+x^2}{x^2}\right)^{1/x}=e^2$, then $f(2)$ is -

(A) 2 (B) 8 (C) 10 (D) 12

MLT242

27. Let $f(x)=\frac{\tan x}{x}$, then the value of $\lim\limits_{x\to 0}\left(\left[f(x)\right]+x^2\right)^{\frac{1}{\{f(x)\}}}$ is equal to (where [.], {.} denotes greatest integer function and fractional part functions respectively) -

(A) e^{-3} (B) e^3 (C) e^2 (D) non-existent

MLT243

28. $\lim\limits_{n\to\infty}\frac{e^n}{\left(1+\frac{1}{n}\right)^{n^2}}$ equals -

(A) 1 (B) $\frac{1}{2}$ (C) e (D) $\sqrt{e}$

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MLT244

29. If $f(x)$ is odd linear polynomial with $f(1) = 1$, then $\lim\limits_{x\to 0}\frac{2^{f(\tan x)}-2^{f(\sin x)}}{x^2 f(\sin x)}$ is

(A) 1 (B) $\ell n2$ (C) $\frac{1}{2}\ell n2$ (D) cos2

MLT245

30. $\lim\limits_{x\to 0}\frac{x(1+a\cos x)-b\sin x}{x^3}$ = 1 then

(A) $a=\frac{-5}{2}$ (B) $a=\frac{-3}{2},b=\frac{-1}{2}$

(C) $a=\frac{-3}{2},b=\frac{-5}{2}$ (D) $a=\frac{-5}{2},b=\frac{-3}{2}$

MLT246

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Let $f(x)=\begin{Bmatrix}\sin x;\text{where } x=\text{integer}\\ 0\ ;\ \text{otherwise}\end{Bmatrix}: g(x)=\begin{Bmatrix}x^2+1 & ; & x\neq 0,2\\ 4 & ; & x=0\\ 5 & ; & x=2\end{Bmatrix}$, then

(A) $\lim\limits_{x\to 0} g(f(x))=4$ (B) $\lim\limits_{x\to 0} f(g(x))=0$

(C) $\lim\limits_{x\to 1} f(g(x))=0$ (D) $\lim\limits_{x\to 1} g(f(x))=5$

MLT055

2. Let $f(x)=\begin{cases}1+\dfrac{2x}{a}, & 0\le x<1\\ ax, & 1\le x<2\end{cases}$, if $\lim\limits_{x\to 1} f(x)$ exists, then value(s) of a is :

(A) 1 (B) –1 (C) 2 (D) –2

MLT056

3. If $A=\lim\limits_{x\to 0}\dfrac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}$ and $B=\lim\limits_{x\to 0}\dfrac{[|x|]}{x}$, then (where [.] denotes greatest integer function)-

(A) $A=1$ (B) A does not exist

(C) $B=0$ (D) $B=1$

MLT058

4. The value(s) of 'n' for which $\lim\limits_{x\to 1}\dfrac{e^{x-1}-x}{(x-1)^n}$ exists is/are -

(A) 1 (B) 2 (C) 3 (D) 4

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MLT060

5. $\lim\limits_{x\to c} f(x)$ does not exist when LHL ≠ RHL (where $[x]$ is the step-up function, $\{x\}$ is the fractional part function of x & sgn(x) denotes signum function), then-

(A) $f(x)=\big[[x]\big]-[2x-1]; c=3$ (B) $f(x)=[x]-x, c=1$

(C) $f(x)=\{x\}^2-\{-x\}^2, c=0$ (D) $f(x)=\dfrac{\tan(\text{sgn}\,x)}{\text{sgn}\,x}, c=0$

MLT062

6. If $f(x)=\dfrac{\cos 2-\cos 2x}{x^2-|x|}$, then

(A) $\lim\limits_{x\to -1} f(x)=2\sin 2$ (B) $\lim\limits_{x\to 1} f(x)=2\sin 2$

(C) $\lim\limits_{x\to -1} f(x)=2\cos 2$ (D) $\lim\limits_{x\to 1} f(x)=2\cos 2$

MLT063

7. Which of the following limits equal to $\dfrac{1}{2}$

(A) $\lim\limits_{n\to\infty}\left(\dfrac{1}{1.3}+\dfrac{1}{3.5}+\ldots+\dfrac{1}{(2n-1)(2n+1)}\right)$ (B) $\lim\limits_{x\to\infty}\left[\dfrac{3x^2}{2x+1}-\dfrac{(2x-1)(3x^2+x+2)}{4x^2}\right]$

(C) $\lim\limits_{n\to\infty}\dfrac{1}{n^2}\ (1+2+3+\ldots\ldots+n)$ (D) $\lim\limits_{n\to\infty}\dfrac{(n+2)!+(n+1)!}{(n+2)!-(n+1)!}$

MLT247

For More Join: @IITJEE_Advanced

8. Which of the following limit tends to unity ?

(A) $\lim\limits_{x\to 0}\dfrac{1-\cos x+2\sin x-\sin^3 x-x^2+3x^4}{\tan^3 x-6\sin^2 x+x-5x^3}$ (B) $\lim\limits_{x\to\infty}\dfrac{x}{[x]}$

(C) $\lim\limits_{x\to\infty}\dfrac{1}{\left(\sqrt{x+\sqrt{x+\sqrt{x}}}-\sqrt{x}\right)}$ (D) $\lim\limits_{x\to\infty}\left(\dfrac{\sqrt{x}}{\sqrt{x+\sqrt{x+\sqrt{x}}}}\right)$

MLT248

9. Let $f:\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)\to\mathbb{R}$, $f(x)=\begin{cases}\lim\limits_{n\to\infty}\left(\dfrac{(\tan x)^{2n}+x^2}{\sin^2 x+(\tan x)^{2n}}\right) & ;\ x\neq 0\\ 1 & ;\ x=0\end{cases}$, $n\in\mathbb{N}$. Which of the following holds good ?

(A) $f\left(-\dfrac{\pi^-}{4}\right)=f\left(\dfrac{\pi^+}{4}\right)$ (B) $f\left(-\dfrac{\pi^-}{4}\right)=f\left(-\dfrac{\pi^+}{4}\right)$

(C) $f\left(\dfrac{\pi^-}{4}\right)=f\left(\dfrac{\pi^+}{4}\right)$ (D) $f(0^+)=f(0)=f(0^-)$

MLT249

10. Let $f(x)=\begin{cases}\dfrac{\tan^2\{x\}}{x^2-[x]^2} & \text{for } x>0\\ 1 & \text{for } x=0\\ \sqrt{\{x\}\cot\{x\}} & \text{for } x<0\end{cases}$, where $[x]$ is the step up function and $\{x\}$ is the fractional part function of x, then-

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(A) $\lim\limits_{x\to 0^+} f(x)=1$ (B) $\lim\limits_{x\to 0^-} f(x)=1$

(C) $\cot^{-1}\left(\lim\limits_{x\to 0^-} f(x)\right)^2=1$ (D) $\lim\limits_{x\to 0^+} f(x)=2$

MLT250

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 11 to 13

Consider two functions $f(x)=\lim\limits_{n\to\infty}\left(\cos\dfrac{x}{\sqrt{n}}\right)^n$ and $g(x)=-x^{4b}$,

Where $b=\lim\limits_{x\to\infty}\left(\sqrt{x^2+x+1}-\sqrt{x^2+1}\right)$, then

11. $f(x)$ is

(A) e^{-x^2} (B) $e^{\frac{-x^2}{2}}$ (C) e^{x^2} (D) $e^{\frac{x^2}{2}}$

MLT064

12. $g(x)$ is

(A) $-x^2$ (B) x^2 (C) x^4 (D) $-x^4$

MLT065

13. Number of solutions of equation $f(x)+g(x)=0$ is

(A) 2 (B) 4 (C) 0 (D) 1

MLT066

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Paragraph for Question No. 14 to 16

Consider $f(x) = \dfrac{\sin x + ae^{x} + be^{-x} + c\ \ell n(1+x)}{x^3}$, where a, b, c are real numbers.

14. If $\lim\limits_{x\to 0^+} f(x)$ is finite, then the value of $a - b + c$ is

MLT067

15. If $\lim\limits_{x\to 0^+} f(x) = \ell$ (finite), then the value of ℓ is

MLT068

16. Using the values of a, b, c as found above, the value of $\lim\limits_{x\to 0^+} x\, f(x)$ is

MLT069

MATCHING LIST TYPE QUESTION

17. $\lim\limits_{x\to 0} f(x)$, where $f(x)$ is as in LIST-I, is

	List–I		**List–II**
(P)	$f(x) = \dfrac{\tan[e^2]x^2 - \tan[-e^2]x^2}{\sin^2 x}$ where [.] represents greatest integer function.	(1)	$\dfrac{\sqrt{2}}{8}$
(Q)	$f(x) = \left[\left(\min(t^2+4t+6)\right)\dfrac{\sin x}{x}\right]$ where [.] represents greatest integer function.	(2)	15
(R)	$f(x) = \dfrac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$	(3)	1
(S)	$f(x) = \dfrac{\sqrt{2} - \sqrt{1+\cos x}}{\sin^2 x}$	(4)	$\dfrac{1}{2}$

(A) (P) → (3), (Q) → (2), (R) → (4), (S) → (1)

(B) (P) → (2), (Q) → (3), (R) → (4), (S) → (1)

(C) (P) → (2), (Q) → (3), (R) → (1), (S) → (4)

(D) (P) → (2), (Q) → (4), (R) → (3), (S) → (1)

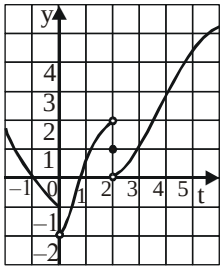
MLT070

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MATRIX MATCH TYPE QUESTION

18. For the function $g(t)$ whose graph is given, match the entries of column-I to column-II



	Column-I		Column-II	
(P)	$\lim_{t\to0^+} g(t) + \lim_{t\to2^-} g(t)$	(1)	$\lim_{t\to2^+} g(t)$	**MLT071**
(Q)	$\lim_{t\to0^-} g(t) + g(2)$	(2)	does not exist	**MLT072**
(R)	$\lim_{t\to0} g(t)$	(3)	0	**MLT073**
(S)	$\lim_{t\to2} g(t)$	(4)	$\lim_{t\to4} g(t)$	**MLT074**

19.

	Column-I		Column-II	
(P)	$\lim_{n\to\infty} n\sin\left(\frac{\pi}{4n}\right)\cos\left(\frac{\pi}{4n}\right)$ is equal to	(1)	0	**MLT075**
(Q)	$\lim_{x\to0} \frac{\sin x^\circ}{x}$ is equal to	(2)	$\frac{1}{2}$	**MLT076**
(R)	$\lim_{x\to0}\left(\frac{1}{\sin x} - \frac{1}{\tan x}\right)$ is equal to	(3)	$\frac{\pi}{4}$	**MLT077**
(S)	$\lim_{x\to\pi/2} \frac{1+\cos 2x}{(\pi-2x)^2}$ is equal to	(4)	$\frac{\pi}{180}$	**MLT078**

20.

	Column-I		Column-II	
(P)	$\lim_{x\to\infty} \frac{2^x}{2^x+1}$ can be equal to	(1)	2	**MLT079**
(Q)	$\lim_{x\to2} \frac{2\sin(e^{x-2}-1)}{\log(x-1)}$ is equal to	(2)	1	**MLT080**
(R)	$\lim_{x\to e} \frac{(\ell n\, x-1)e}{x-e}$ is equal to	(3)	$\frac{1}{2}$	**MLT081**
(S)	$\lim_{x\to0} \frac{x(5^x-1)}{(1-\cos x)4\ell n5}$ is equal to	(4)	0	**MLT082**

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EXERCISE (S)

1. $\lim\limits_{x\to\infty}\dfrac{(x+1)^{10}+(x+2)^{10}+....+(x+100)^{10}}{x^{10}+10^{10}}$ is equal to

MLT021

2. Let $P_n=\prod\limits_{k=2}^{n}\left(1-\dfrac{1}{{}^{k+1}C_2}\right)$. If $\lim\limits_{n\to\infty}P_n$ can be expressed as lowest rational in the form $\dfrac{a}{b}$, then value of $(a+b)$ is

MLT022

3. $\lim\limits_{x\to0}\left(\left[\dfrac{-5\sin x}{x}\right]+\left[\dfrac{6\sin x}{x}\right]\right)$ (where [.] denotes greatest integer function) is equal to –

MLT023

4. Let $f(x)=\dfrac{\sin\{x\}}{x^2+ax+b}$. If $f(5^+)$ & $f(3^+)$ exists finitely and are not zero, then the value of $(a+b)$ is (where {.} represents fractional part function) -

MLT024

5. $\lim\limits_{n\to\infty}\left(4^n+5^n\right)^{\frac{1}{n}}$ is equal to

MLT025

6. $\underset{x\to1}{Lim}\dfrac{x^2-x.\ln x+\ln x-1}{x-1}$

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MLT026

7. $\lim\limits_{x\to0}\ [\ell n(1+\sin^2x).\cot(\ell n^2\ (1+x))]$

MLT027

8. If $\underset{x\to\infty}{Lim}\dfrac{a(2x^3-x^2)+b(x^3+5x^2-1)-c(3x^3+x^2)}{a(5x^4-x)-bx^4+c(4x^4+1)+2x^2+5x}=1$, then the value of $(a+b+c)$ can be expressed in the lowest form as $\dfrac{p}{q}$. Find the value of $(p+q)$.

MLT028

9. $\lim\limits_{x\to0}\dfrac{2^x-1}{(1+x)^{1/2}-1}=ln\,k$, where k is equal to

MLT029

10. $L=\lim\limits_{x\to\infty}x^2\left(\sqrt{\dfrac{x+2}{x}}-\sqrt[3]{\dfrac{x+3}{x}}\right)$ then $2L$ equal to

MLT030

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EXERCISE - JEE (Main) PYQ

1. For each $t \in R$, let $[t]$ be the greatest integer less than or equal to t. Then

$\lim\limits_{x\to 0+} x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+....+\left[\frac{15}{x}\right]\right)$ **[JEE (Main) 2018]**

(1) is equal to 15. (2) is equal to 120.

(3) does not exist (in R). (4) is equal to 0.

MLT038

2. $\lim\limits_{y\to 0}\frac{\sqrt{1+\sqrt{1+y^4}}-\sqrt{2}}{y^4}$ **[JEE (Main) 2019]**

(1) exists and equals $\frac{1}{4\sqrt{2}}$ (2) does not exist

(3) exists and equals $\frac{1}{2\sqrt{2}}$ (4) exists and equals $\frac{1}{2\sqrt{2}\left(\sqrt{2}+1\right)}$

MLT039

3. For each $x \in R$, let $[x]$ be the greatest integer less than or equal to x. Then $\lim\limits_{x\to 0^-}\frac{x\left([x]+|x|\right)\sin[x]}{|x|}$ is equal to **[JEE (Main) 2019]**

(1) – sin1 (2) 0 (3) 1 (4) sin1

MLT040

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4. For each $t \in R$, let $[t]$ be the greatest integer less than or equal to t.

Then, $\lim\limits_{x\to 1+}\frac{(1-|x|+\sin|1-x|)\sin\left(\frac{\pi}{2}[1-x]\right)}{|1-x|[1-x]}$ **[JEE (Main) 2019]**

(1) equals –1 (2) equals 1 (3) does not exist (4) equals 0

MLT041

5. Let $[x]$ denote the greatest integer less than or equal to x.

Then $\lim\limits_{x\to 0}\frac{\tan(\pi\sin^2 x)+\left(|x|-\sin\left(x[x]\right)\right)^2}{x^2}$ **[JEE (Main) 2019]**

(1) equals π (2) equals 0 (3) equals π + 1 (4) does not exist

MLT042

6. $\lim\limits_{x\to \pi/4}\frac{\cot^3 x-\tan x}{\cos\left(x+\pi/4\right)}$ is : **[JEE (Main) 2019]**

(1) 4 (2) $8\sqrt{2}$ (3) 8 (4) $4\sqrt{2}$

MLT043

7. $\lim\limits_{x\to 1-}\frac{\sqrt{\pi}-\sqrt{2\sin^{-1}x}}{\sqrt{1-x}}$ equal to: **[JEE (Main) 2019]**

(1) $\frac{1}{\sqrt{2\pi}}$ (2) $\sqrt{\frac{\pi}{2}}$ (3) $\sqrt{\frac{2}{\pi}}$ (4) $\sqrt{\pi}$

MLT044

For More Join: @IITJEE_Advanced

8. $\lim\limits_{x\to 0}\dfrac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$ equals: **[JEE (Main) 2019]**

(1) $2\sqrt{2}$ (2) $4\sqrt{2}$ (3) $\sqrt{2}$ (4) 4

MLT045

9. If $\lim\limits_{x\to 1}\dfrac{x^4-1}{x-1}=\lim\limits_{x\to k}\dfrac{x^3-k^3}{x^2-k^2}$, then k is : **[JEE (Main) 2019]**

(1) $\dfrac{3}{8}$ (2) $\dfrac{3}{2}$ (3) $\dfrac{4}{3}$ (4) $\dfrac{8}{3}$

MLT046

10. If $\lim\limits_{x\to 1}\dfrac{x^2-ax+b}{x-1}=5$, then $a+b$ is equal to :- **[JEE (Main) 2019]**

(1) –7 (2) – 4 (3) 5 (4) 1

MLT047

11. $\lim\limits_{x\to 2}\dfrac{3^x+3^{3-x}-12}{3^{-x/2}-3^{1-x}}$ is equal to. **[JEE (Main) 2020]**

MLT048

12. $\lim\limits_{x\to 0}\left(\dfrac{3x^2+2}{7x^2+2}\right)^{\frac{1}{x^2}}$ is equal to **[JEE (Main) 2020]**

(1) $\dfrac{1}{e}$ (2) e^2 (3) e (4) $\dfrac{1}{e^2}$

MLT049

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13. If $\lim\limits_{x\to 1}\dfrac{x+x^2+x^3+...+x^n-n}{x-1}=820, (n\in \mathrm{N})$ then the value of n is equal to. **[JEE (Main) 2020]**

MLT050

14. $\lim\limits_{x\to 0}\left(\tan\left(\dfrac{\pi}{4}+x\right)\right)^{1/x}$ is equal to : **[JEE (Main) 2020]**

(1) 2 (2) e (3) 1 (4) e^2

MLT051

15. $\lim\limits_{n\to\infty}\tan\left\{\sum\limits_{r=1}^{n}\tan^{-1}\left(\dfrac{1}{1+r+r^2}\right)\right\}$ is equal to. **[JEE (Main) 2021]**

MLT052

16. The value of $\lim\limits_{x\to 0^+}\dfrac{\cos^{-1}(x-[x]^2)\cdot\sin^{-1}(x-[x]^2)}{x-x^3}$, where $[x]$ denotes the greatest integer $\le x$ is :

[JEE (Main) 2021]

(1) π (2) 0 (3) $\dfrac{\pi}{4}$ (4) $\dfrac{\pi}{2}$

MLT097

17. If $\lim\limits_{x\to\infty}\left(\sqrt{x^2-x+1}-ax\right)=b$, then the ordered pair (a, b) is: **[JEE (Main) 2021]**

(1) $\left(1,\dfrac{1}{2}\right)$ (2) $\left(1,-\dfrac{1}{2}\right)$ (3) $\left(-1,\dfrac{1}{2}\right)$ (4) $\left(-1,-\dfrac{1}{2}\right)$

MLT098

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18. $\lim\limits_{x\to 0}\dfrac{\sin^2\left(\pi\cos^4 x\right)}{x^4}$ is equal to : **[JEE (Main) 2021]**

(1) π^2 (2) $2\pi^2$ (3) $4\pi^2$ (4) 4π

MLT099

19. $\lim\limits_{x\to\frac{\pi}{2}}\left(\tan^2 x\left(\left(2\sin^2 x+3\sin x+4\right)^{\frac{1}{2}}-\left(\sin^2 x+6\sin x+2\right)^{\frac{1}{2}}\right)\right)$ is equal to **[JEE (Main) 2022]**

(1) $\dfrac{1}{12}$ (2) $-\dfrac{1}{18}$ (3) $-\dfrac{1}{12}$ (4) $-\dfrac{1}{6}$

MLT100

20. $\lim\limits_{x\to 0}\dfrac{\cos\left(\sin x\right)-\cos x}{x^4}$ is equal to : **[JEE (Main) 2022]**

(1) $\dfrac{1}{3}$ (2) $\dfrac{1}{4}$ (3) $\dfrac{1}{6}$ (4) $\dfrac{1}{12}$

MLT101

21. $\lim\limits_{x\to\frac{1}{\sqrt{2}}}\dfrac{\sin(\cos^{-1}x)-x}{1-\tan(\cos^{-1}x)}$ is equal to : **[JEE (Main) 2022]**

(1) $\sqrt{2}$ (2) $-\sqrt{2}$ (3) $\dfrac{1}{\sqrt{2}}$ (4) $-\dfrac{1}{\sqrt{2}}$

MLT102

22. $\lim\limits_{x\to\infty}\dfrac{\left(\sqrt{3x+1}+\sqrt{3x-1}\right)^6+\left(\sqrt{3x+1}-\sqrt{3x-1}\right)^6}{\left(x+\sqrt{x^2-1}\right)^6+\left(x-\sqrt{x^2-1}\right)^6}x^3$

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[JEE (Main) 2023]

(1) is equal to 9 (2) is equal to 27 (3) does not exist (4) is equal to $\dfrac{27}{2}$

MLT103

23. The value of $\lim\limits_{n\to\infty}\dfrac{1+2-3+4+5-6+\ldots+(3n-2)+(3n-1)-3n}{\sqrt{2n^4+4n+3}-\sqrt{n^4+5n+4}}$ is : **[JEE (Main) 2023]**

(1) $\dfrac{\sqrt{2}+1}{2}$ (2) $3(\sqrt{2}+1)$ (3) $\dfrac{3}{2}(\sqrt{2}+1)$ (4) $\dfrac{3}{2\sqrt{2}}$

MLT104

24. The set of all values of a for which $\lim\limits_{x\to a}\left([x-5]-[2x+2]\right)=0$, where [x] denotes the greater integer less than or equal to x is equal to

[JEE (Main) 2023]

(1) (– 7.5, –6.5) (2) (– 7.5, –6.5] (3) [– 7.5, –6.5] (4) [– 7.5, –6.5)

MLT105

25. $\lim\limits_{t\to 0}\left(1^{\frac{1}{\sin^2 t}}+2^{\frac{1}{\sin^2 t}}+\ldots.+n^{\frac{1}{\sin^2 t}}\right)^{\sin^2 t}$ is equal to **[JEE (Main) 2023]**

(1) n^2+n (2) n (3) $\dfrac{n(n+1)}{2}$ (4) n^2

MLT106

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26. If $\lim\limits_{x\to 0}\dfrac{e^{ax}-\cos(bx)-\dfrac{cxe^{-cx}}{2}}{1-\cos(2x)}=17$, then $5a^2 + b^2$ is equal to

[JEE (Main) 2023]

(1) 72 (2) 76 (3) 68 (4) 64

MLT107

27. If $\lim\limits_{x\to 1}\dfrac{(5x+1)^{1/3}-(x+5)^{1/3}}{(2x+3)^{1/2}-(x+4)^{1/2}}=\dfrac{m\sqrt{5}}{n(2n)^{2/3}}$, where $gcd(m, n) = 1$, then $8m + 12n$ is equal to ____

[JEE (Main) 2024]

MLT146

28. Let $a > 0$ be a root of the equation $2x^2 + x - 2 = 0$. If $\lim\limits_{x\to\frac{1}{a}}\dfrac{16\left(1-\cos(2+x-2x^2)\right)}{(1-ax^2)}=\alpha+\beta\sqrt{17}$, where $\alpha, \beta \in Z$ then $\alpha + \beta$ is equal to _____.

[JEE (Main) 2024]

MLT147

29. $\lim\limits_{n\to\infty}\dfrac{(1^2-1)(n-1)+(2^2-2)(n-2)+.....+\left((n-1)^2-(n-1)\right)\cdot 1}{(1^3+2^3+.....+n^3)-(1^2+2^2+.....+n^2)}$ is equal to:

[JEE (Main) 2024]

(1) $\frac{2}{3}$ (2) $\frac{1}{3}$ (3) $\frac{3}{4}$ (4) $\frac{1}{2}$

MLT148

30. If $\alpha=\lim\limits_{x\to 0^+}\left(\dfrac{e^{\sqrt{\tan x}}-e^{\sqrt{x}}}{\sqrt{\tan x}-\sqrt{x}}\right)$ and $\beta=\lim\limits_{x\to 0}(1+\sin x)^{\frac{1}{2}\cot x}$ are the roots of the quadratic equation $ax^2 + bx - \sqrt{e} = 0$, then $12\log_e(a + b)$ is equal to _____.

[JEE (Main) 2024]

MLT149

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EXERCISE - JEE (Advanced) PYQ

1. Let $L=\lim\limits_{x\to 0}\dfrac{a-\sqrt{a^2-x^2}-\dfrac{x^2}{4}}{x^4}$, $a>0$. If L is finite, then - **[IIT JEE 2009]**

(A) $a=2$ (B) $a=1$ (C) $L=\dfrac{1}{64}$ (D) $L=\dfrac{1}{32}$

MLT083

2. If $\lim\limits_{x\to 0}\left[1+x\ell n(1+b^2)\right]^{\frac{1}{x}}=2b\sin^2\theta, b>0$ and $\theta\in(-\pi,\pi]$, then the value of θ is- **[IIT JEE 2011]**

(A) $\pm\dfrac{\pi}{4}$ (B) $\pm\dfrac{\pi}{3}$ (C) $\pm\dfrac{\pi}{6}$ (D) $\pm\dfrac{\pi}{2}$

MLT084

3. If $\lim\limits_{x\to\infty}\left(\dfrac{x^2+x+1}{x+1}-ax-b\right)=4$, then - **[IIT JEE 2012]**

(A) $a=1, b=4$ (B) $a=1, b=-4$
(C) $a=2, b=-3$ (D) $a=2, b=3$

MLT085

4. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $\left(\sqrt[3]{1+a}-1\right)x^2+\left(\sqrt{1+a}-1\right)x+\left(\sqrt[6]{1+a}-1\right)=0$ where $a>-1$. Then $\lim\limits_{a\to 0^+}\alpha(a)$ and $\lim\limits_{a\to 0^+}\beta(a)$ are **[IIT JEE 2012]**

(A) $-\dfrac{5}{2}$ and 1 (B) $-\dfrac{1}{2}$ and -1 (C) $-\dfrac{7}{2}$ and 2 (D) $-\dfrac{9}{2}$ and 3

MLT086

5. The largest value of the non-negative integer a for which $\lim\limits_{x\to 1}\left\{\dfrac{-ax+\sin(x-1)+a}{x+\sin(x-1)-1}\right\}^{\frac{1-x}{1-\sqrt{x}}}=\dfrac{1}{4}$ is **[JEE (Advanced) 2014]**

MLT087

6. Let m and n be two positive integers greater than 1. If $\lim\limits_{\alpha\to 0}\left(\dfrac{e^{\cos(\alpha^n)}-e}{\alpha^m}\right)=-\left(\dfrac{e}{2}\right)$ then the value of $\dfrac{m}{n}$ is **[JEE (Advanced) 2015]**

MLT088

7. Let $\alpha, \beta\in\mathbb{R}$ be such that $\lim\limits_{x\to 0}\dfrac{x^2\sin(\beta x)}{\alpha x-\sin x}=1$. Then $6(\alpha+\beta)$ equals **[JEE (Advanced) 2016]**

MLT089

8. Let $f(x)=\dfrac{1-x(1+|1-x|)}{|1-x|}\cos\left(\dfrac{1}{1-x}\right)$ for $x\neq 1$. Then **[JEE (Advanced) 2017]**

(C) $\lim\limits_{x\to 1^-}f(x)=0$ (D) $\lim\limits_{x\to 1^+}f(x)=0$
(A) $\lim\limits_{x\to 1^+}f(x)$ does not exist (B) $\lim\limits_{x\to 1^-}f(x)$ does not exist

MLT090

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9. For any positive integer n, define $f_n:(0,\infty)\to\mathbb{R}$ as

$$f_n(x)=\sum_{j=1}^{n}\tan^{-1}\left(\frac{1}{1+(x+j)(x+j-1)}\right) \text{ for all } x\in(0,\infty).$$

(Here, the inverse trigonometric function $\tan^{-1}x$ assume values in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$)

Then, which of the following statement(s) is (are) TRUE ? **[JEE (Advanced) 2018]**

(A) $\sum_{j=1}^{5}\tan^2\left(f_j(0)\right)=55$

(B) $\sum_{j=1}^{10}\left(1+f_j'(0)\right)\sec^2\left(f_j(0)\right)=10$

(C) For any fixed positive integer n, $\lim_{x\to\infty}\tan\left(f_n(x)\right)=\frac{1}{n}$

(D) For any fixed positive integer n, $\lim_{x\to\infty}\sec^2\left(f_n(x)\right)=1$

MLT091

10. Let $F:\mathbb{R}\to\mathbb{R}$ be a function. We say that f has

PROPERTY 1 if $\lim_{h\to0}\frac{f(h)-f(0)}{\sqrt{|h|}}$ exists and is finite, and

PROPERTY 2 if $\lim_{h\to0}\frac{f(h)-f(0)}{h^2}$ exists and is finite.

Then which of the following options is/are correct ? **[JEE (Advanced) 2019]**

(A) $f(x)=x|x|$ has PROPERTY 2 (B) $f(x)=x^{2/3}$ has PROPERTY 1

(C) $f(x)=\sin x$ has PROPERTY 2 (D) $f(x)=|x|$ has PROPERTY 1

MLT092

11. Let e denote the base of the natural logarithm. The value of the real number a for which the right-hand limit $\lim_{x\to0^+}\frac{(1-x)^{\frac{1}{x}}-e^{-1}}{x^a}$ is equal to a nonzero real number, is. **[JEE (Advanced) 2020]**

MLT093

12. The value of the limit $\lim_{x\to\frac{\pi}{2}}\frac{4\sqrt{2}(\sin3x+\sin x)}{\left(2\sin2x\sin\frac{3x}{2}+\cos\frac{5x}{2}\right)-\left(\sqrt{2}+\sqrt{2}\cos2x+\cos\frac{3x}{2}\right)}$ is

[JEE (Advanced) 2020]

MLT094

13. Let α be a positive real number. Let $f:\mathbb{R}\to\mathbb{R}$ and $g:(\alpha,\infty)\to\mathbb{R}$ be the functions defined by

$f(x)=\sin\left(\frac{\pi x}{12}\right)$ and $g(x)=\frac{2\log_e\left(\sqrt{x}-\sqrt{\alpha}\right)}{\log_e\left(e^{\sqrt{x}}-e^{\sqrt{\alpha}}\right)}$. Then the value of $\lim_{x\to\alpha^+}f(g(x))$ is _____.

[JEE (Advanced) 2022]

MLT095

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14. If $\beta=\lim\limits_{x\to 0}\dfrac{e^{x^3}-\left(1-x^3\right)^{\frac{1}{3}}+\left(\left(1-x^2\right)^{\frac{1}{2}}-1\right)\sin x}{x\sin^2 x}$, then the value of 6β is. **[JEE (Advanced) 2022]**

MLT096

15. Let $k\in\mathbb{R}$. If $\lim\limits_{x\to 0^+}(\sin(\sin kx)+\cos x+x)^{\frac{2}{x}}=e^6$, then the value of k is **[JEE (Advanced) 2024]**

(A) 1 (B) 2 (C) 3 (D) 4

MLT150

16. Let S be the set of all $(\alpha,\beta)\in\mathbb{R}\times\mathbb{R}$ such that

$$\lim_{x\to\infty}\frac{\sin(x^2)(\log_e x)^{\alpha}\sin\left(\frac{1}{x^2}\right)}{x^{\alpha\beta}(\log_e(1+x))^{\beta}}=0.$$

Then which of the following is (are) correct ? **[JEE (Advanced) 2024]**

(A) $(-1,3)\in S$ (B) $(-1,1)\in S$ (C) $(1,-1)\in S$ (D) $(1,-2)\in S$

MLT151

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	B	C	B	A	A	C	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	D	A	A	C	D	D	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	A	C	A	D	B	D	C	D

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C	B,C	B,C	A,B	B,C	A,B	A,C	B,D	A,D	A,C
Que.	11	12	13	14	15	16	17	18		
Ans.	B	A	A	–1.00	–0.33	0	B	P → 1,3; Q → 1,3; R → 2; S → 2		
Que.	19				20					
Ans.	P → 3; Q → 4; R → 1; S → 2				P→ 2; Q→1; R→2; S→3					

EXERCISE (S)

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1.	100	**2.**	4	**3.**	0	**4.**	7	**5.**	5
6.	2	**7.**	1	**8.**	167	**9.**	4	**10.**	1

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	1	4	4	3	3	2	4	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	36	4	40	4	1	4	2	3	1	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	2	3	1	2	3	100	170	2	6

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	D	B	B	0	2	7	A,C	D	B,D
Que.	11	12	13	14	15	16				
Ans.	1.00	8	0.50	5	B	B,C				

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Important Notes

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